Longitudinal Data: Homework 1

Question 1:

(a)

(i)Cross-sectional design

$$\hat{\gamma}_a = \bar{Y}_1^A - \bar{Y}_1^B$$

Since there are K/2 participants in each treatment group,

$$\text{Var}(\bar{Y}_1^A) = \frac{\sigma^2}{K/2} = \frac{2\sigma^2}{K},$$

and similarly,

$$\text{Var}(\bar{Y}_1^B) = \frac{2\sigma^2}{K}.$$

The two treatment groups are independent, so

$$\begin{aligned}
\text{Var}(\hat{\gamma}_a) &= \text{Var}(\bar{Y}_1^A - \bar{Y}_1^B) \\
&= \text{Var}(\bar{Y}_1^A) + \text{Var}(\bar{Y}_1^B) \\
&= \frac{2\sigma^2}{K} + \frac{2\sigma^2}{K} \\
&= \boxed{\frac{4\sigma^2}{K}}.
\end{aligned}$$

(ii) Longitudinal comparison of change from baseline

$$\hat{\gamma}_b = (\bar{Y}_1^A - \bar{Y}_0^A) - (\bar{Y}_1^B - \bar{Y}_0^B)$$

For one participant, the variance of the change from baseline is

$$\begin{aligned}
\text{Var}(Y_1 - Y_0) &= \text{Var}(Y_1) + \text{Var}(Y_0) - 2\,\text{Cov}(Y_1, Y_0) \\
&= \sigma^2 + \sigma^2 - 2\rho\sigma^2 \\
&= 2\sigma^2(1-\rho).
\end{aligned}$$

Each treatment group has K/2 participants, so

$$\text{Var}(\bar{Y}_1^A - \bar{Y}_0^A) = \frac{2\sigma^2(1-\rho)}{K/2} = \frac{4\sigma^2(1-\rho)}{K}.$$

The two treatment groups are independent, so

$$\begin{aligned}\mathrm{Var}(\hat{\gamma}_b) &= \frac{4\sigma^2(1-\rho)}{K} + \frac{4\sigma^2(1-\rho)}{K} \\ &= \boxed{\frac{8\sigma^2(1-\rho)}{K}}.\end{aligned}$$

(iii) Crossover design.

$$\hat{\gamma}_c = \bar{Y}_1^A - \bar{Y}_2^B$$

Since the same K participants are measured under both treatments,

$$\begin{aligned}\mathrm{Var}(Y_1^A - Y_2^B) &= \mathrm{Var}(Y_1^A) + \mathrm{Var}(Y_2^B) - 2\,\mathrm{Cov}(Y_1^A, Y_2^B) \\ &= \sigma^2 + \sigma^2 - 2\rho\sigma^2 \\ &= 2\sigma^2(1-\rho).\end{aligned}$$

Since there are K participants,

$$\boxed{\mathrm{Var}(\hat{\gamma}_c) = \frac{2\sigma^2(1-\rho)}{K}}.$$

(iv) Longitudinal comparison of averages.

$$\hat{\gamma}_d = \bar{Y}^A - \bar{Y}^B,$$

where

$$\bar{Y}^{tx} = \frac{Y_1^{tx} + Y_2^{tx}}{2}.$$

For one participant,

$$\begin{aligned}\mathrm{Var}\left(\frac{Y_1 + Y_2}{2}\right) &= \frac{1}{4}\left[\mathrm{Var}(Y_1) + \mathrm{Var}(Y_2) + 2\,\mathrm{Cov}(Y_1, Y_2)\right] \\ &= \frac{1}{4}\left[\sigma^2 + \sigma^2 + 2\rho\sigma^2\right] \\ &= \frac{\sigma^2(1+\rho)}{2}.\end{aligned}$$

Each treatment group has K/2 participants, so

$$\mathrm{Var}(\bar{Y}^A) = \frac{\sigma^2(1+\rho)/2}{K/2} = \frac{\sigma^2(1+\rho)}{K}.$$

Similarly,

$$\mathrm{Var}(\bar{Y}^B) = \frac{\sigma^2(1+\rho)}{K}.$$

Therefore,

$$\begin{aligned}\mathrm{Var}(\hat{\gamma}_d) &= \mathrm{Var}(\bar{Y}^A) + \mathrm{Var}(\bar{Y}^B) \\ &= \frac{\sigma^2(1+\rho)}{K} + \frac{\sigma^2(1+\rho)}{K} \\ &= \boxed{\frac{2\sigma^2(1+\rho)}{K}}.\end{aligned}$$

(b)

There are 300000/500=600 available measurements. The cross-sectional design enrolls 600 people (300 per treatment). Each longitudinal design enrolls 300 people: 150 per treatment in designs (ii) and (iv), or all 300 receiving both treatments in design (iii).

The table reports $\mathrm{Var}(\widehat{\gamma})/\sigma^2$; multiply each entry by σ^2 for the variance itself.

```
variance_table <- function(K_cross, K_long) {
  rho <- c(.2, .5, .8)
  data.frame(
    rho = rho,
    `Cross-sectional` = rep(4 / K_cross, 3),
    `Change from baseline` = 8 * (1 - rho) / K_long,
    Crossover = 2 * (1 - rho) / K_long,
    `Comparison of average` = 2 * (1 + rho) / K_long,
    check.names = FALSE
  )
}
var_b <- variance_table(600, 300)
knitr::kable(var_b, digits = 6)
```

rho	Cross-sectional	Change from baseline	Crossover	Comparison of average
0.2	0.006667	0.021333	0.005333	0.008
0.5	0.006667	0.013333	0.003333	0.010
0.8	0.006667	0.005333	0.001333	0.012

At $\rho = 0.2$, crossover minimizes variance ($0.005333\sigma^2$), followed by cross-sectional, averages, and changes. At $\rho = 0.5$, crossover again wins ($0.003333\sigma^2$); cross-sectional is next, then averages, then changes. At $\rho = 0.8$, crossover wins ($0.001333\sigma^2$), followed by changes, cross-sectional, and averages. Under this budget, change scores improve on the cross-sectional design only when $\rho > 0.75$.

(c)

A cross-sectional participant costs 250+250=500, so 600 people (300 each) can be enrolled. A longitudinal participant costs 250+2(250)=750, so 400 people can be enrolled: 200 per treatment in designs (ii) and (iv), or all 400 receiving both treatments in design (iii).

```
var_c <- variance_table(600, 400)
knitr::kable(var_c, digits = 6)
```

rho	Cross-sectional	Change from baseline	Crossover	Comparison of average
0.2	0.006667	0.016	0.0040	0.0060
0.5	0.006667	0.010	0.0025	0.0075
0.8	0.006667	0.004	0.0010	0.0090

```
stopifnot(600 * 500 == 300000, 300 * 2 * 500 == 300000,
          400 * (250 + 2 * 250) == 300000,
          all(var_b$Crossover < var_b$`Cross-sectional`),
          all(var_c$Crossover < var_c$`Cross-sectional`))
```

At $\rho = 0.2$, crossover minimizes variance ($0.004000\sigma^2$), followed by averages, cross-sectional, and changes. At $\rho = 0.5$, crossover wins ($0.002500\sigma^2$), followed by cross-sectional, averages, and changes. At $\rho = 0.8$, crossover wins ($0.001000\sigma^2$), followed by changes, cross-sectional, and averages. Changes improve on cross-sectional sampling when $\rho > 2/3$; averages improve on it when $\rho < 1/3$. Since the goal is to minimize variance, the crossover design would be chosen for all three values of ρ.

Question 2

(a)

First, I checked the number of girls, the number of visits, and missing values.

```
topeka <- topeka[order(topeka$id, topeka$age), ]
topeka$FEV1 <- exp(topeka$log.FEV1)
visits <- table(topeka$id)

c(girls = length(visits), observations = nrow(topeka),
  one_visit = sum(visits == 1), missing_values = sum(is.na(topeka)),
  duplicate_visits = sum(duplicated(topeka[c("id", "age")])))
```

```
##            girls     observations        one_visit   missing_values
##              300             1994               48                0
## duplicate_visits
##                0
```

```
# A short summary table
variables <- topeka[c("age", "height", "FEV1", "log.FEV1")]
descriptive <- data.frame(
  Mean = sapply(variables, mean),
  SD = sapply(variables, sd),
  Minimum = sapply(variables, min),
  Median = sapply(variables, median),
  Maximum = sapply(variables, max))
knitr::kable(descriptive, digits = 2)
```

	Mean	SD	Minimum	Median	Maximum
age	12.57	3.32	6.43	12.60	18.69
height	1.50	0.15	1.11	1.54	1.79

	Mean	SD	Minimum	Median	Maximum
FEV1	2.38	0.74	0.50	2.38	4.93
log.FEV1	0.82	0.33	-0.69	0.87	1.60

```
summary(as.numeric(visits))
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##   1.000   3.000   7.000   6.647  10.000  12.000
```

There are 300 girls and 1,994 observations. Each girl has between 1 and 12 visits, and 48 girls have only one visit. There are no missing values or duplicate ID-age pairs. Since the girls have different numbers of visits, I would use a method that allows unequal follow-up. Girls with one visit can help describe lung function levels, but not changes within a girl.

Mean structure

```
par(mfrow = c(1, 2), mar = c(4, 4, 1, 1))
plot(topeka$age, topeka$FEV1, pch = 16, cex = .4,
     col = "gray65", xlab = "Age (years)", ylab = "FEV1 (L)")
lines(lowess(topeka$age, topeka$FEV1), col = "blue", lwd = 2)
plot(topeka$height, topeka$FEV1, pch = 16, cex = .4,
     col = "gray65", xlab = "Height (m)", ylab = "FEV1 (L)")
lines(lowess(topeka$height, topeka$FEV1), col = "blue", lwd = 2)
```

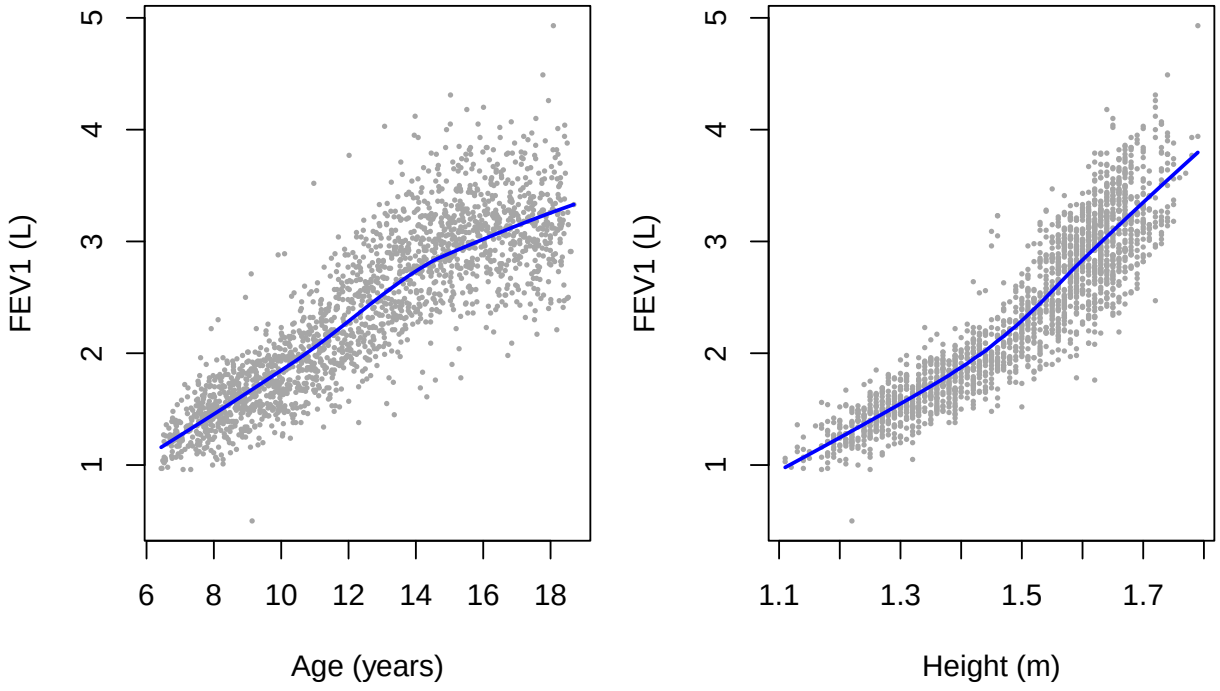


Figure 1: FEV1 by age and height, with smooth curves.

```r
par(mfrow = c(1, 1))
```

FEV1 generally increases with age and height. The age curve starts to level off in the later teenage years, so a straight line may not describe the whole age range well. I would consider a curved age effect and include height when modeling lung function. Since age and height are related, their effects should be interpreted together.

```r
set.seed(8157)
selected_ids <- sample(unique(topeka$id), 20)
plot_data <- subset(topeka, id %in% selected_ids)
ggplot(plot_data, aes(age, log.FEV1, group = id)) +
  geom_line(alpha = .5) + geom_point(size = .7) +
  labs(x = "Age (years)", y = "Log FEV1")
```

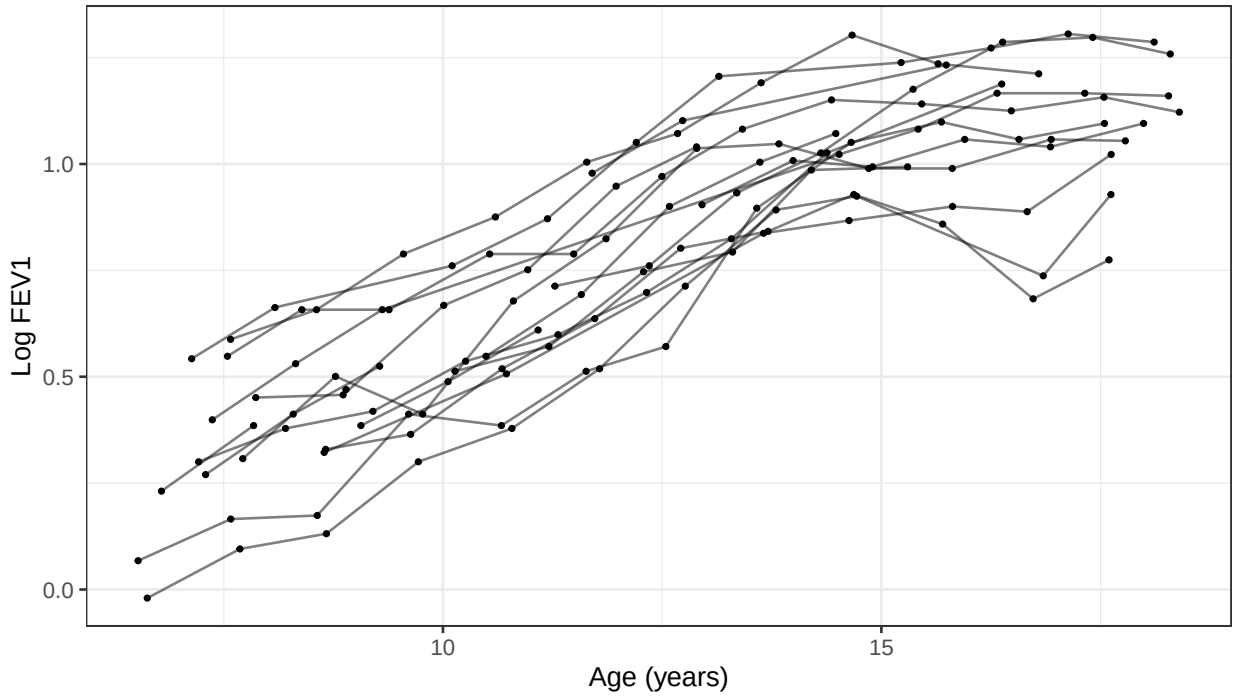


Figure 2: Log FEV1 over time for 20 randomly selected girls.

Most girls show increasing lung function, but their starting levels and patterns of growth differ. Some also have decreases between visits. This suggests allowing girls to have different starting levels and possibly different growth rates.

Outliers

I used a simple curved mean model to look for observations that are unusual for a girl's age and height.

```r
mean_model <- lm(log.FEV1 ~ age + I(age^2) + height, data = topeka)
topeka$residual <- residuals(mean_model)
topeka$std_residual <- rstandard(mean_model)

# Show the five largest residuals in absolute value.
unusual <- order(abs(topeka$std_residual), decreasing = TRUE)[1:5]
```

```r
knitr::kable(topeka[unusual,
  c("id", "age", "height", "FEV1", "std_residual")],
  digits = 3, row.names = FALSE)
```

id	age	height	FEV1	std_residual
197	9.136	1.22	0.50	-8.546
246	14.579	1.62	1.76	-4.193
118	13.202	1.59	1.78	-3.435
32	8.274	1.32	1.05	-3.380
131	18.259	1.72	2.47	-3.326

Girl 197 has a particularly low FEV1 of about 0.50 L at age 9.14. I would check this measurement against the original record before deciding whether it is an error. I would not remove an observation only because it is unusual. Also, a negative log FEV1 is possible when FEV1 is below 1 L.

Dependence structure

To look at dependence, I compared residuals from consecutive observed visits within each girl. Removing the age and height trend first helps avoid mistaking average growth for correlation.

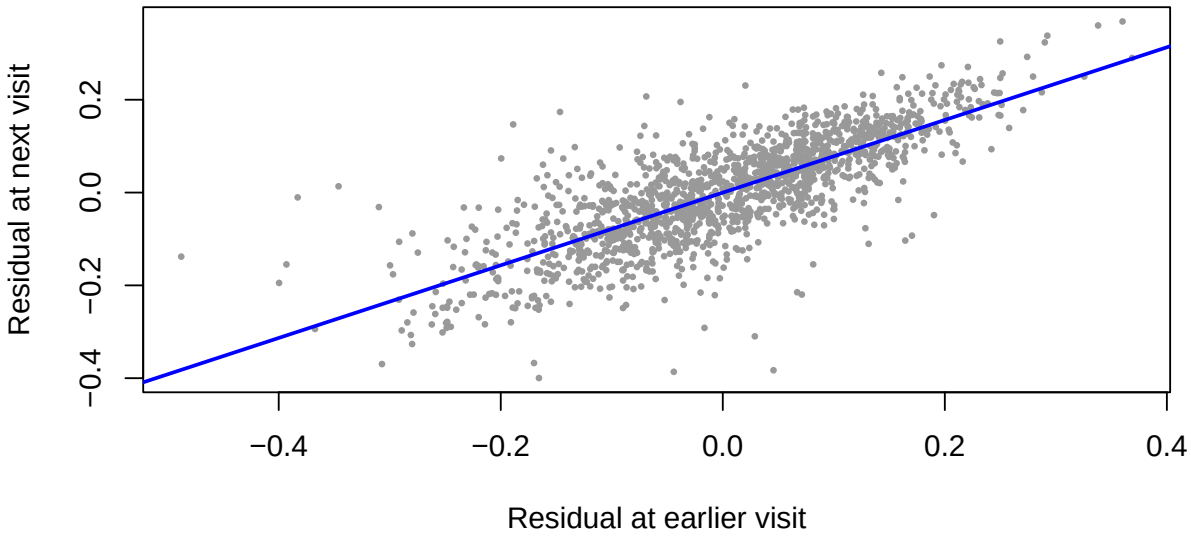
```r
previous <- c()
current <- c()
time_gap <- c()

for (girl in unique(topeka$id)) {
  d <- subset(topeka, id == girl)
  n <- nrow(d)
  if (n >= 2) {
    previous <- c(previous, d$residual[1:(n - 1)])
    current <- c(current, d$residual[2:n])
    time_gap <- c(time_gap, diff(d$age))
  }
}

# Restrict to approximately annual pairs for a comparable time gap.
annual <- time_gap >= .5 & time_gap <= 1.5
annual_cor <- cor(previous[annual], current[annual])
data.frame(Pairs = sum(annual), Residual_correlation = round(annual_cor, 3))
```

```
##   Pairs Residual_correlation
## 1  1566                0.785
```

```r
plot(previous[annual], current[annual], pch = 16, cex = .5,
     col = "gray60", xlab = "Residual at earlier visit",
     ylab = "Residual at next visit")
abline(lm(current[annual] ~ previous[annual]), col = "blue", lwd = 2)
```



The residual correlation for visits about one year apart is 0.79. This shows that measurements from the same girl are positively related even after adjusting for age and height. I would account for within-girl correlation, for example with a random-intercept model, rather than treating all visits as independent. This is a descriptive correlation; the pairs themselves are not independent, and it does not tell us the correlation at every time gap.

(b)

1. How does lung function change as girls get older, and when does growth slow down?
2. Do girls differ in their starting lung function and rates of growth?
3. Is an increase in a girl's height associated with an increase in her lung function?
4. Does a girl's earlier lung function help predict her later lung function?

(c)

With one measurement per girl, we could compare FEV1 between girls of different ages or heights. However, an age difference between two girls is not necessarily the same as a change within one girl. Interpreting the cross-sectional age pattern as growth would require assuming that girls of different ages are otherwise comparable, including their birth cohorts and selection into the study. Baseline ages also cover less of adolescence than all follow-up visits.

We could describe differences in lung function levels, but we could not estimate individual growth rates or their variation. We could study the association between height and FEV1 across girls, but not the association between changes in height and FEV1 within a girl. Finally, without later measurements, we could not estimate within-girl correlation or check how well baseline lung function predicts future lung function.

Question 3

(a) Baseline Table 1

```
macs <- macs[order(macs$id, macs$time), ]
baseline <- data.frame()
followup <- data.frame()

for (person in unique(macs$id)) {
  d <- subset(macs, id == person)
  before <- subset(d, time >= -.5 & time < 0)
  after <- subset(d, time > 0)

  if (nrow(before) > 0 && nrow(after) > 0) {
    b <- before[which.max(before$time), ]
    baseline <- rbind(baseline, b)
    b$time <- 0
    followup <- rbind(followup, b, after)
  }
}
nrow(baseline)
```

```
## [1] 266
```

```
# Continuous baseline variables
x <- baseline[c("age", "partners", "cesd", "cd4")]
table1 <- data.frame(
  Mean = sapply(x, mean),
  SD = sapply(x, sd),
  Median = sapply(x, median),
  Q1 = sapply(x, quantile, probs = .25),
  Q3 = sapply(x, quantile, probs = .75))
knitr::kable(table1, digits = 1,
             caption = "Baseline characteristics (N = 266).")
```

Table 5: Baseline characteristics (N = 266).

	Mean	SD	Median	Q1	Q3
age	2.3	7.4	1.3	-2.9	6.3
partners	6.7	3.5	8.0	3.0	10.0
cesd	10.0	10.0	7.0	2.2	14.0
cd4	998.9	422.4	932.5	696.2	1220.8

```
# Categorical baseline variables
packs_table <- as.data.frame(table(baseline$packs))
names(packs_table) <- c("Packs code", "N")
packs_table$Percent <- 100 * packs_table$N / nrow(baseline)
knitr::kable(packs_table, digits = 1)
```

Packs code	N	Percent
0	159	59.8
1	11	4.1
2	16	6.0
3	55	20.7
4	25	9.4

```
drug_table <- as.data.frame(table(baseline$drug))
names(drug_table) <- c("Drug code", "N")
drug_table$Percent <- 100 * drug_table$N / nrow(baseline)
knitr::kable(drug_table, digits = 1)
```

Drug code	N	Percent
0	48	18
1	218	82

The final sample contains 266 participants, with no missing baseline covariates. The mean baseline CD4 count is 998.9 cells/mm^3.

(b) Two-stage analysis

Stage 1

For each participant, I fit the model

$$\text{CD4}_{ki} = a_k + b_k \text{time}_{ki} + e_{ki}.$$

The intercept estimates the CD4 level at time zero, and the slope estimates the change per year. I chose a linear model because it is easy to interpret and some participants only have two measurements.

```
individual_results <- data.frame()

for (person in unique(followup$id)) {
  d <- subset(followup, id == person)
  fit <- lm(cd4 ~ time, data = d)
  individual_results <- rbind(individual_results, data.frame(
    id = person,
    intercept = unname(coef(fit)[1]),
    slope = unname(coef(fit)[2]),
    visits = nrow(d),
    last_time = max(d$time)))
}

knitr::kable(data.frame(
  Quantity = c("Intercept", "Annual change"),
  Mean = c(mean(individual_results$intercept), mean(individual_results$slope)),
  SD = c(sd(individual_results$intercept), sd(individual_results$slope)),
  Median = c(median(individual_results$intercept), median(individual_results$slope))),
  digits = 1)
```

Quantity	Mean	SD	Median
Intercept	905.8	330.3	846.8
Annual change	-293.4	504.6	-141.7

The mean slope is -293.4 cells/mm^3 per year, and the median is -141.7. Both suggest decreasing CD4 counts. The mean is more negative because some participants have very steep estimated declines.

```
set.seed(8157)
selected_ids <- sample(unique(followup$id), 20)
ggplot(subset(followup, id %in% selected_ids),
       aes(time, cd4, group = id)) +
  geom_line(alpha = .5) + geom_point(size = .7) +
  labs(x = "Years since seroconversion", y = "CD4 count (cells/mm^3)")
```

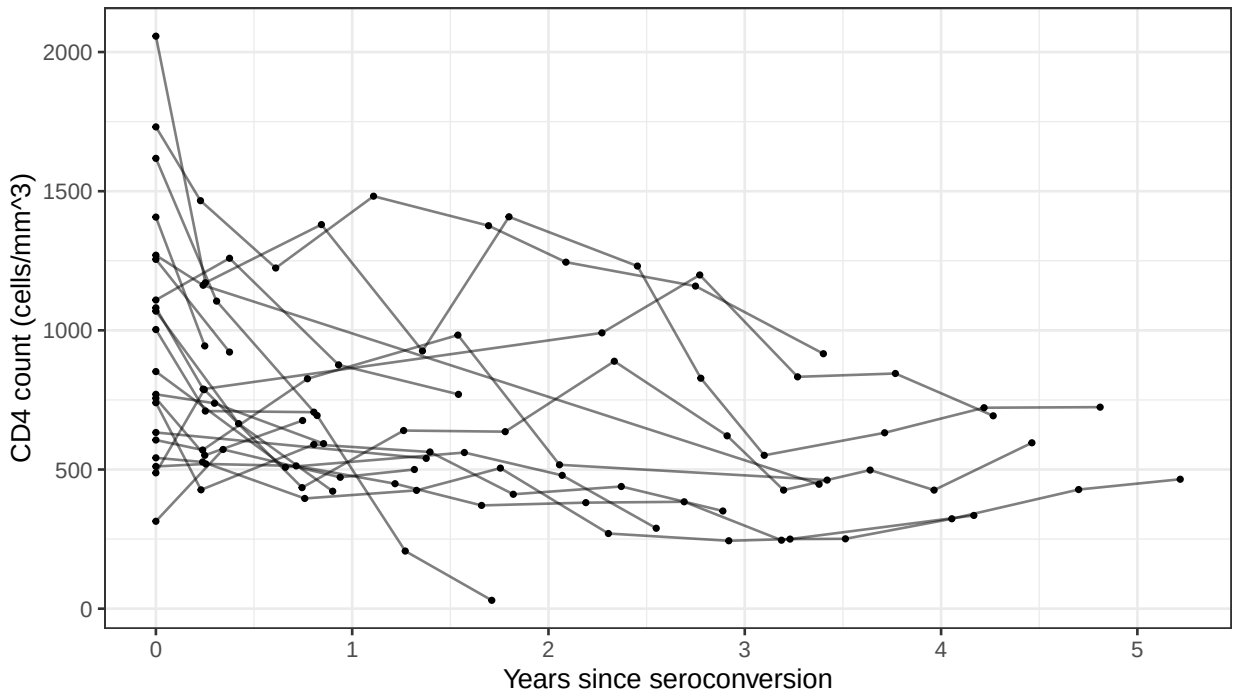


Figure 3: CD4 trajectories for 20 randomly selected participants.

Stage 2

Next, I used baseline age, smoking category, drug code, partners, and CES-D to predict the estimated intercepts and slopes. Each participant contributes one row to these regressions. I did not include baseline CD4 as a predictor because it was already used to estimate the intercept and slope.

```
results <- merge(baseline, individual_results, by = "id")
results$packs <- factor(results$packs)
results$drug <- factor(results$drug)

intercept_model <- lm(intercept ~ age + packs + drug + partners + cesd,
                      data = results)
slope_model <- lm(slope ~ age + packs + drug + partners + cesd,
```

```
                  data = results)

# Ordinary least squares estimates and confidence intervals
intercept_table <- cbind(summary(intercept_model)$coefficients,
                         confint(intercept_model))
slope_table <- cbind(summary(slope_model)$coefficients,
                     confint(slope_model))
intercept_table <- as.data.frame(intercept_table)
slope_table <- as.data.frame(slope_table)
intercept_table[["Pr(>|t|)"]] <- ifelse(
  intercept_table[["Pr(>|t|)"]] < .001, "<0.001",
  sprintf("%.3f", intercept_table[["Pr(>|t|)"]]))
slope_table[["Pr(>|t|)"]] <- ifelse(
  slope_table[["Pr(>|t|)"]] < .001, "<0.001",
  sprintf("%.3f", slope_table[["Pr(>|t|)"]]))
knitr::kable(intercept_table, digits = 3,
             caption = "Baseline predictors of the fitted CD4 intercept.")
```

Table 9: Baseline predictors of the fitted CD4 intercept.

	Estimate	Std. Error	t value	Pr(>|t|)	2.5 %	97.5 %
(Intercept)	798.096	57.956	13.771	<0.001	683.967	912.224
age	2.654	2.635	1.007	0.315	-2.536	7.843
packs1	-76.160	94.054	-0.810	0.419	-261.375	109.055
packs2	152.466	80.219	1.901	0.058	-5.504	310.436
packs3	319.687	47.503	6.730	<0.001	226.142	413.232
packs4	143.637	66.454	2.161	0.032	12.774	274.500
drug1	113.708	49.923	2.278	0.024	15.398	212.019
partners	-8.375	5.436	-1.541	0.125	-19.080	2.330
cesd	-2.101	1.882	-1.116	0.265	-5.806	1.604

```
knitr::kable(slope_table, digits = 3,
             caption = "Baseline predictors of the fitted annual CD4 change.")
```

Table 10: Baseline predictors of the fitted annual CD4 change.

	Estimate	Std. Error	t value	Pr(>|t|)	2.5 %	97.5 %
(Intercept)	-272.297	97.139	-2.803	0.005	-463.587	-81.007
age	-3.094	4.417	-0.700	0.484	-11.792	5.604
packs1	221.629	157.644	1.406	0.161	-88.809	532.066
packs2	-120.886	134.454	-0.899	0.369	-385.658	143.886
packs3	-117.525	79.620	-1.476	0.141	-274.315	39.265
packs4	118.338	111.382	1.062	0.289	-101.000	337.677
drug1	-62.248	83.676	-0.744	0.458	-227.026	102.530
partners	2.671	9.111	0.293	0.770	-15.271	20.613
cesd	3.041	3.154	0.964	0.336	-3.169	9.251

In the intercept model, smoking codes 3 and 4 and drug code 1 are associated with higher fitted CD4 levels than their reference groups, holding the other variables fixed ($p<0.05$). For example, the estimate for

smoking code 3 is about 319.7 cells/mm^3 higher than code 0. Age, partners, CES-D, and the other smoking categories do not have clear associations at the 0.05 level in this model.

In the slope model, none of the individual baseline covariates has a p-value below 0.05 using the usual least squares standard errors. The model explains only 3.5% of the variation in slopes. Thus, these baseline variables do not explain much of the difference in estimated decline between participants.

```
# Check how much short follow-up affects the average slope.
longer_followup <- subset(results, visits >= 3 & last_time >= 1)
knitr::kable(data.frame(
  Sample = c("All participants", "At least 3 visits and 1 year of follow-up"),
  N = c(nrow(results), nrow(longer_followup)),
  Mean_slope = c(mean(results$slope), mean(longer_followup$slope)),
  Median_slope = c(median(results$slope), median(longer_followup$slope))),
  digits = 1)
```

Sample	N	Mean_slope	Median_slope
All participants	266	-293.4	-141.7
At least 3 visits and 1 year of follow-up	214	-155.6	-120.4

The average decline was smaller among participants who had more follow-up, although the median still showed a decline. This suggests that participants with shorter follow-up may have had a strong influence on the average decline. We kept all 266 participants in the main analysis because excluding participants with shorter follow-up would also change the group being studied.

Summary for non-biostatisticians CD4 counts generally decreased after seroconversion. About 92.5% of participants showed a decrease over time, with a typical decline of about 142 cells/mm^3 per year. Some characteristics measured at the start of the study were related to CD4 counts around the time of seroconversion, but they did not explain much of the difference in how quickly CD4 counts changed over time. The average rate of decline should be interpreted cautiously because participants with shorter follow-up periods sometimes had unusually large estimated changes. These findings show relationships between characteristics and CD4 counts; they do not show that any particular characteristic caused CD4 counts to change.